

Bug Report - Login Failure (401 Unauthorized)

MERN Stack - User Authentication Issue

Date: July 10, 2025

Summary

Between July 3 and July 10, 2025, users were unable to log in even with the correct password. The backend consistently returned a 401 Unauthorized error with the message 'Credenciais inválidas.'

Problem

Passwords were stored in hashed format, but login verification failed. Initial suspicion pointed to issues with `bcrypt.compare` or data mismatch.

Investigation Steps

- Verified frontend was sending the correct fields (email, password).
- Confirmed passwords were hashed in MongoDB.
- Checked `userController.js`: passwords were hashed before saving.
- Inspected `User.js`: found a `pre('save')` hook also hashing passwords.
- Concluded that passwords were being double-hashed (`bcrypt(bcrypt(password))`).

Solution

- Removed the `pre('save')` hook from the Mongoose schema (`User.js`).
- Maintained single-point hashing inside the `userController.js` register function.
- Cleaned test users with invalid hashes and tested login with a new account.

Result

Login worked successfully after registering a new user. `bcrypt.compare` returned true and token was issued.

Timeline

- Issue detected: July 3, 2025
- Resolved: July 10, 2025
- Duration: ~7 days

Developer

Allan Rodrigo, with technical guidance and debugging support via ChatGPT.